

QRAFT TECHNOLOGIES, LONG-TERM EQUITY TEAM, 2024

LLM News Sentiment as a Trading Factor

Turning daily stock news into probability-based sentiment scores and a weekly long/short portfolio — the LLM as a universe selector, not a stock picker.

Juyeong Lee, project lead and AI Quant Researcher

Why trade on news sentiment?

What news adds that financial statements cannot



Market sensitivity

Prices react instantly to earnings, M&A and regulation; news is usually the first trigger.



Leading indicator

Headlines shape short-term investor psychology — buying momentum after good news, sell-offs after bad.



Beyond fundamentals

Statements are lagged and backward-looking; news is real-time and forward-looking.



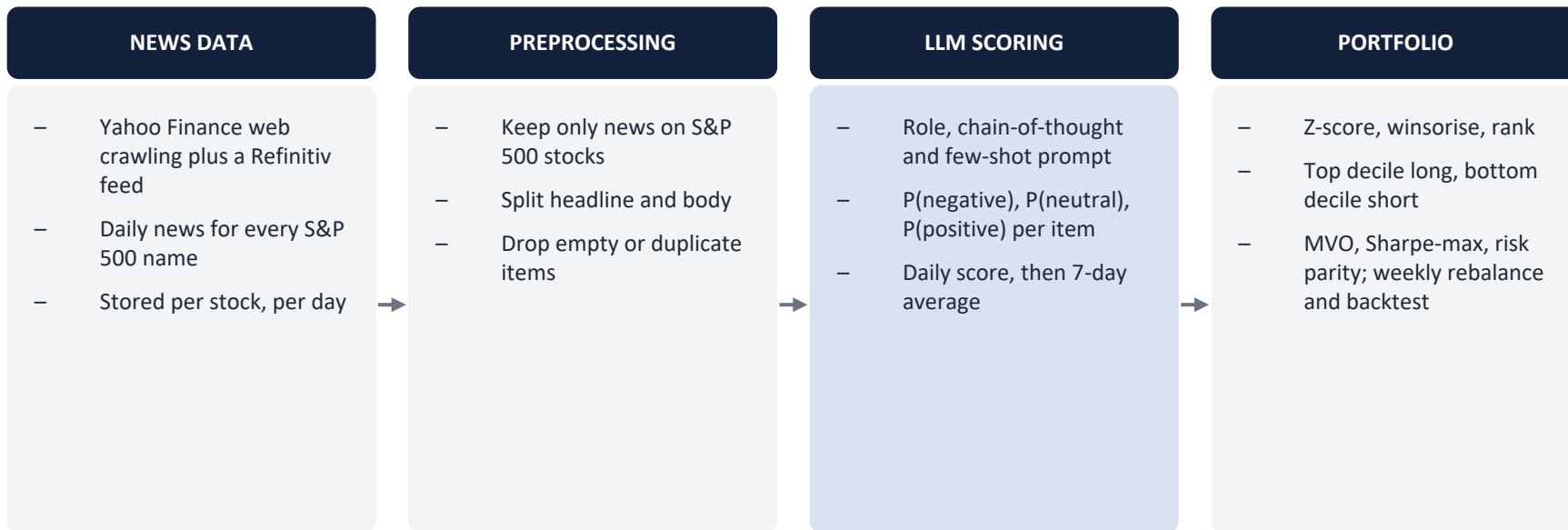
Qualitative signals

Management credibility, brand perception, regulatory risk and innovation — hard to quantify any other way.

Combining fundamentals with sentiment improves predictive power — the question is how to make sentiment a clean, numeric factor.

End-to-end pipeline

From raw news to a weekly long/short book



Crawling replaced an expensive vendor subscription while keeping full coverage of the index.

Prompt design: probabilities, not adjectives

The model is asked for three probabilities that sum to one, grounded only in the article

SYSTEM PROMPT

"You are a financial sentiment analysis assistant. Classify the sentiment towards the company by assigning probabilities for positive, neutral and negative, each between 0 and 1, where the three values sum to exactly 1. Use only the content of the news — do not invent facts ..."

PROMPT TECHNIQUES

Role prompting

fixes the analyst persona and output format

Chain of thought

reasons through the article before scoring

Few-shot examples

anchors the scale so scores are comparable

Date	Stock	Negative	Neutral	Positive
1/1	APPLE	0.1	0.2	0.7
1/1	TESLA	0.2	0.3	0.5
1/1	META	0.1	0.8	0.1
1/1	NIKE	0.5	0.4	0.1

Output per news item, per stock, per day. Probabilities are easier to combine and normalise than labels, and expose the model's uncertainty.

From probabilities to a daily and weekly score

Neutral is ignored; the negative probability counts against the stock

Stock	Neg	Neu	Pos
APPLE	0.1	0.2	0.7
TESLA	0.2	0.3	0.5
META	0.1	0.8	0.1
NIKE	0.5	0.4	0.1



$$\text{score} = P(\text{pos}) - P(\text{neg})$$

neutral ignored



Stock	Daily score
APPLE	$0.7 - 0.1 = 0.6$
TESLA	$0.5 - 0.2 = 0.3$
META	$0.1 - 0.1 = 0.0$
NIKE	$0.1 - 0.5 = -0.4$

DAILY → WEEKLY: 7-DAY AVERAGE PER STOCK

Week of	APPLE	TESLA	META
1/1	0.5	0.4	0.5
1/8	0.6	0.8	0.4
1/15	0.6	0.3	0.0

Loss-aversion variant

A resampling variant gives the negative probability a heavier weight, reflecting that investors react more to bad news than to good.

Portfolio construction

Weekly scores become a ranked long/short book

1

Universe: S&P 500 names with news coverage (US-listed, above \$5, no pending M&A)

2

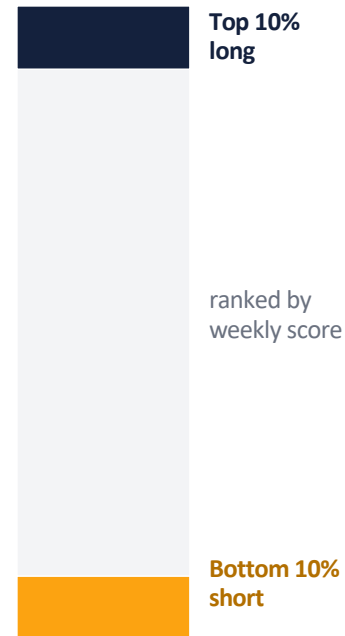
Weekly score $\rightarrow$ cross-sectional z-score, winsorised at ± 3

3

Rank: top decile long, bottom decile short, score-weighted

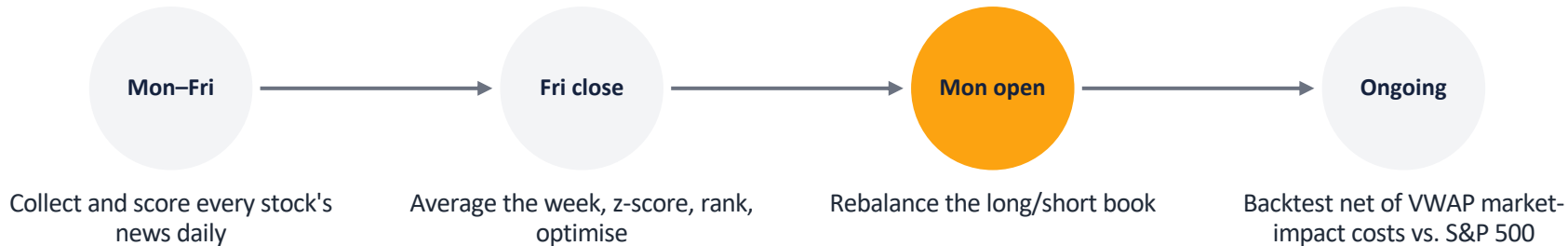
4

Optimiser variants: Markowitz mean-variance, Sharpe maximisation, cardinality constraints, risk parity



Weekly cycle and backtest

Run at week end, apply at the next week's start



Backtest protocol

- Weekly closing prices; 100% turnover assumed
- Universe filters and z-score checks identical to the multi-factor pipeline

Cost model

- Market impact: BB+ model with the order split into 78 five-minute VWAP slices, cross-checked with ITG
- Benchmark: S&P 500 total return

Results and takeaways

Backtest, net of market-impact costs

≈ **15%**

annual return of the sentiment-enhanced strategy

1.2

Sharpe ratio

Outperformed the S&P 500 in backtests; most effective in narrative- and event-driven markets where fundamentals lag the news.

Sentiment score combined with value and growth factors in the multi-factor framework; positions sized with mean-variance optimisation and risk parity.

LLM as universe selector, not stock picker

The model filters the index into a high-potential trading universe; the quant engine sizes positions.

Replicable text pipeline

Crawling plus prompt-engineered scoring brought unstructured text into systematic trading at low cost.

In production

Used as one of the strategies managing the SoftBank fund (fund size confidential).

Takeaways

Ask the LLM for calibrated probabilities, not labels — they normalise, average and rank like any other factor.

Prompt engineering (role, chain of thought, few-shot) is what makes scores comparable across thousands of articles.

Sentiment complements slower fundamental signals: a real-time factor for event-driven markets.

Same discipline as the multi-factor pipeline — universe filters, z-scores, deciles and a realistic cost model.

Python, LangGraph, GPT, SQL, Snowflake

Juyeong Lee, juyeong.lee0839@gmail.com